

BIOMOLECULES

Carbohydrates $\rightarrow$ Hydrates of carbon

• General formula $\rightarrow C_x(H_2O)_y$

Glucose	$C_6H_{12}O_6$	$C_6(H_2O)_6$ ✓
Acetic Acid	$C_2H_4O_2$	$C_2(H_2O)_2$ ✗
Rhamnose	$C_6H_{12}O_5$	$C_6(H_2O)_5H_2$ ✓

$\rightarrow$ Rhamnose, $C_6H_{12}O_5$ is a carbohydrate but does not fit in this definition.

★ Defⁿ: - Optically Active polyhydroxy aldehydes or ketones or the comp. which produce such units on hydrolysis.

Classification of Hydrates

On the basis of Hydrolysis

① Monosaccharide :- which does not hydrolyse further.

eg $\rightarrow$ Glucose, fructose, Galactose etc.

② Oligosaccharide :- produces 2-10 monosac units on hydrolysis

• Disac $\rightarrow$ 2 unit

Sucrose $\xrightarrow{H_2O}$ g + f

Maltose $\xrightarrow{''}$ 2 x g

Lactose $\xrightarrow{''}$ g + gl

• Trisac $\rightarrow$ 3 unit

eg $\rightarrow$ Rabinose

③ Polysaccharides :- gives more than 10 monosac on hydrolysis.

eg $\rightarrow$ starch, cellulose, glycogen, gums etc.

$\rightarrow$ Not sweet in taste $\rightarrow$ Non-sugar.

On the basis of Reducing & non-reducing Sugars

Reducing Sugar

Non-Reducing Sugar

① Tollen test ✓

✗

② Fehling test ✓

✗

③ ~~He~~ Benedict's Reagent ✓

✗

★ All monosacch. like glucose, fructose

★ Disac like, Maltose, lactose etc.

★ All polysacch.

★ Disac like Sucrose.

On the basis of no. of carbons

No. of C	Name	Aldose	Ketose
3c	Triose	Aldo Triose	Keto Triose
4c	Tetrose	Aldo Tetrose	Keto Tetrose
5c	Pentose	Aldo Pentose	Keto Pentose
6c	Hexose	Aldo Hexose	Keto Hexose

MOP OF GLUCOSE

① Sucrose $\xrightarrow{H_3O^+}$ Glucose + fructose
Invert sugar $C_6H_{12}O_6$ $C_6H_{12}O_6$
(+) dextro (+) d (-) l
- (l)

② From Hydrolysis of cellulose & starch

$(C_6H_{10}O_5)_n \xrightarrow{H_3O^+} n C_6H_{12}O_6$
starch or cellulose glucose

★ Properties

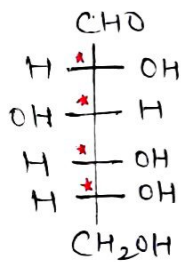
① Glucose also known as dextrose (+) or Aldohexose.

• If last OH in Right side $\rightarrow$ d (+)

• If last OH in left side $\rightarrow$ l (-)

② Structure ($C_6H_{12}O_6$)

Fischer projection formula



Chiral C ($n=4$)

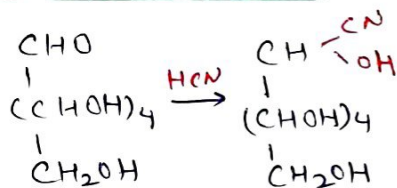
Total stereoisomers = 2^n
 $= 2^4 = 16$

D-glucose

★ Chemical Reactions

① Rxn with HCN

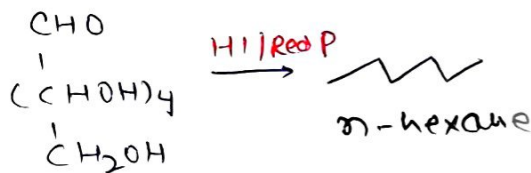
(confirm presence of carbonyl group)



glucose
cyanohydrine

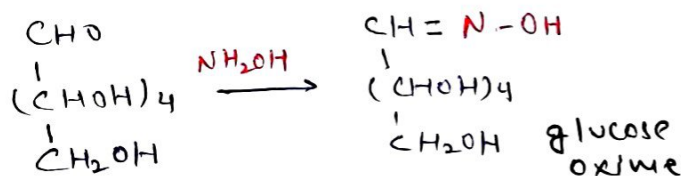
Total chiral carbons = 5
 " stereoisomers = $2^5 = 32$

② Rxn with HI/Red P



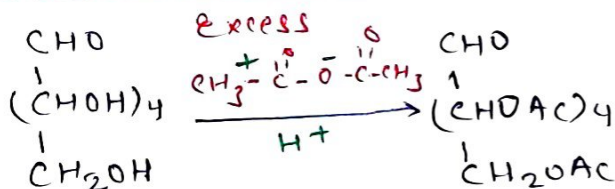
n-hexane

③ Rxn with NH_2OH (hydroxylamine)



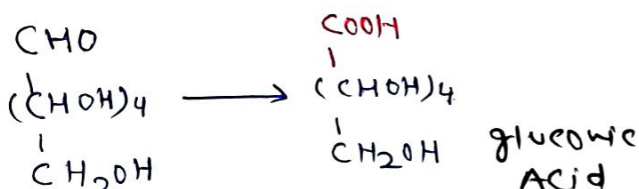
glucose
oxime

④ Rxn with acetic anhydride



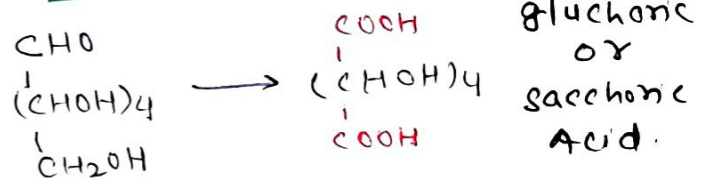
(confirm presence of S-OH group)
 glucose
pentaacetate

⑤ Rxn with $\text{Br}_2 + \text{H}_2\text{O}$



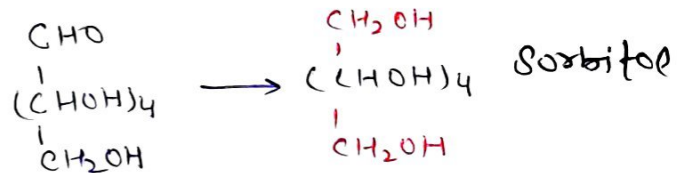
gluconic
Acid

⑥ Rxn with conc. HNO_3



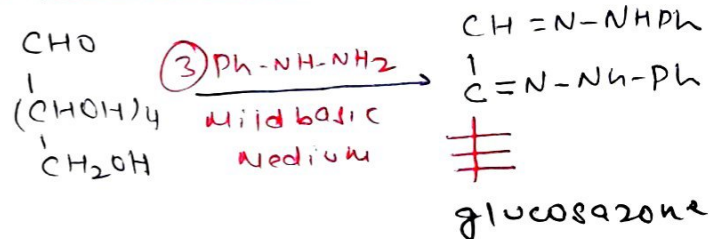
gluconic
or
saccharic
Acid.

⑦ Rxn with $\text{Na(Hg)} + \text{ROH}$ or $\text{NaBH}_4/\text{MeOH}$ or $\text{H}_2 + \text{Catalyst}$



Sorbitol

⑧ Osazone formation



glucosazone

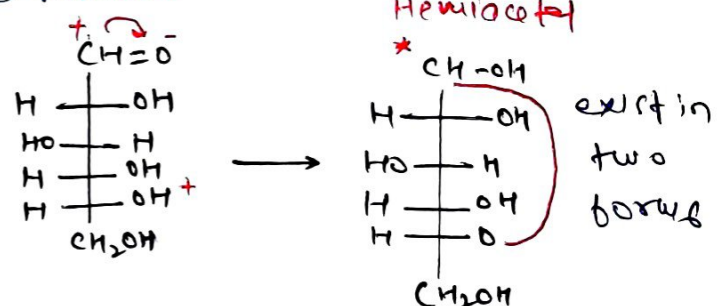
★ Rxn which support cyclic structure

- ① No rxn with NaHSO_3
- ② No rxn with Schiff's test
- ③ No rxn with 2,4 DNP Test
- ④ No rxn with glucose penta acetate with NH_2OH .

★ Glucose exist in 3-different form

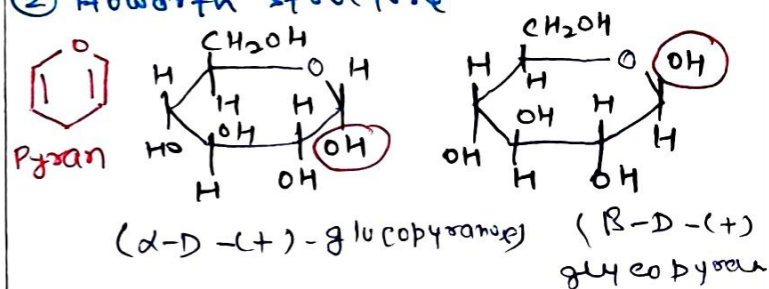
→ (α , β & Fischer)

① fischer



exists in
two
forms

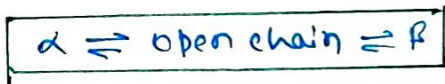
② Haworth structure



(α -D-(+)-glucopyranose) (β -D-(+)-glucopyranose)

★ α & β are anomers (C-1 anomers)
 (Diastereomers)

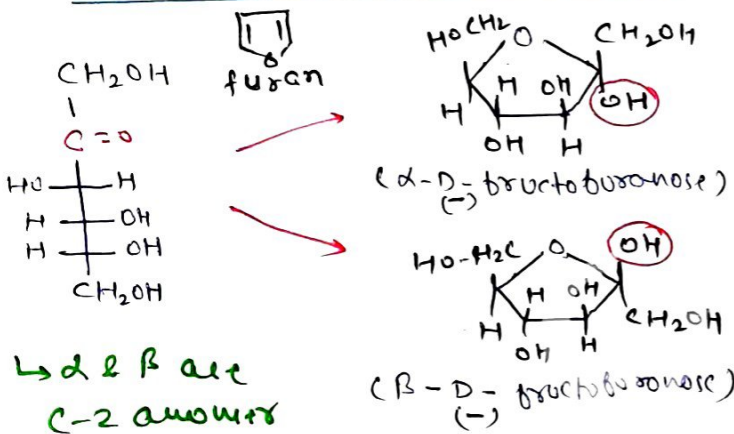
*** Mutarotation** → Inter conversion of α & β in acidic or basic medium is called mutarotation (due to hemiacetal)



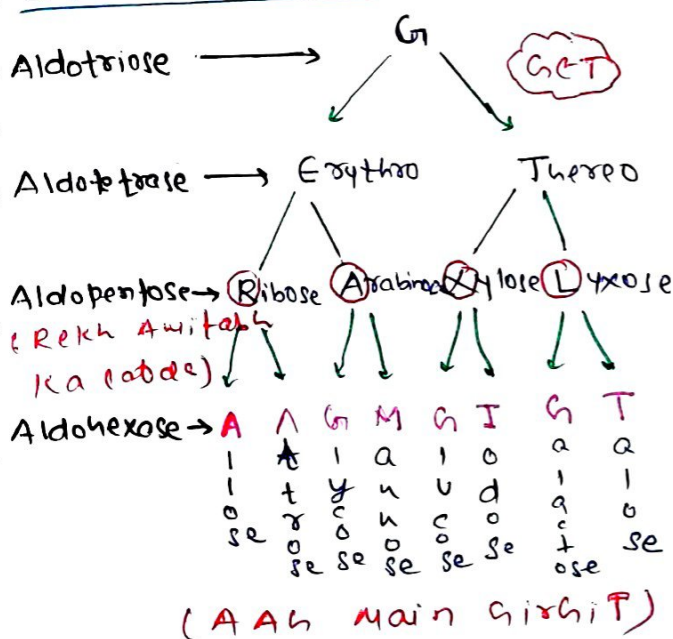
*** OP Points**

- Net optical rotation of glucose = $+52.4^\circ$.
- $\therefore \alpha = 36.1\%$ (sp. rotation = $+112^\circ$)
 $\therefore \beta = 63.8\%$ (sp. rotation = $+19^\circ$)
- α & β are diastereomers or epimers or Anomers (not mirror image)
- L-Glucose is the mirror of D-glucose.

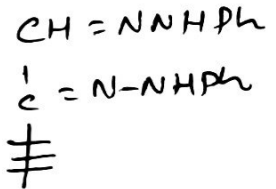
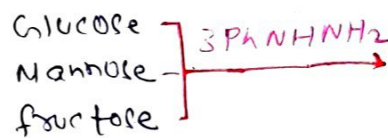
Cyclic structure of fructose



Super trick to remember Monosaccharide



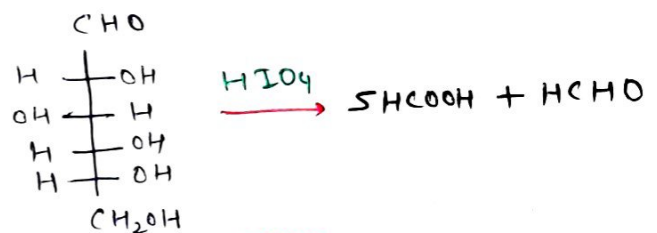
*** Glucose, Mannose & fructose give same osazone.**



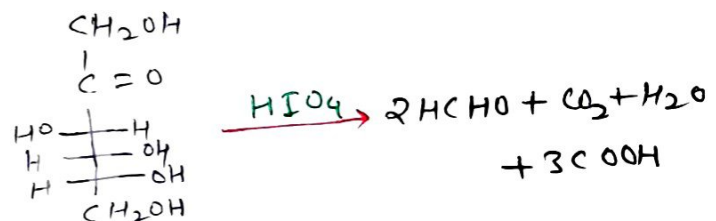
*** Chemical rxn of glucose & fructose**

*** Reaction with HIO₄**

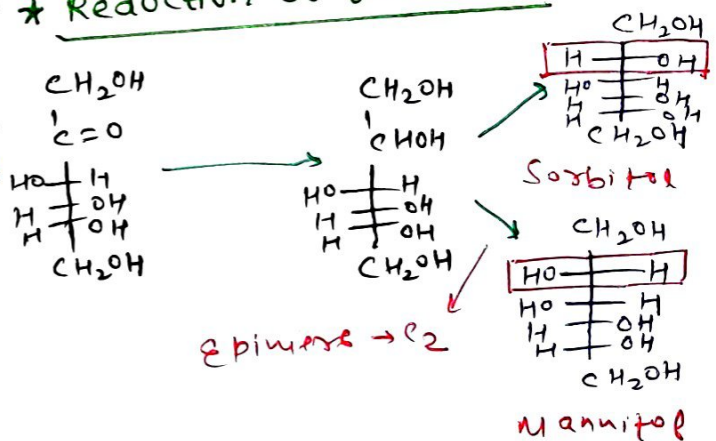
Glucose



Fructose



*** Reduction of fructose**



*** OP POINTS:-**

- Glucose & fructose can be distinguished by bromine water not by T.R, F.R, B.R.
- Glucose, fructose & Mannose give same osazone bcz in osazone formation take place at first two carbon atoms.
- osazone is stable by inter-molecular H-bonding.

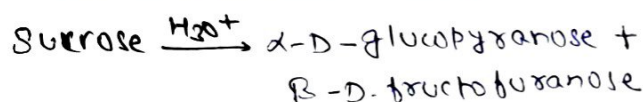
*** Epimers:-** (Same MF, diastereomer)

- glucose & Mannose C-2 epimer
- glucose & galactose C-4 epimer

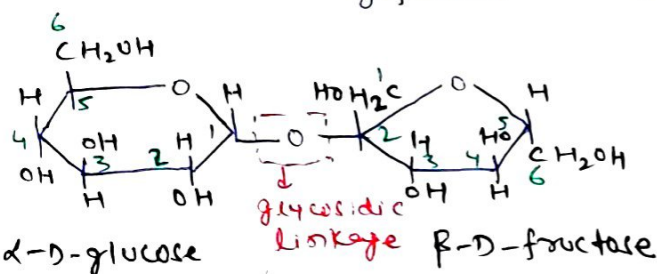
- ④ Glucose / fructose is converted to ethyl alcohol by fermentation in presence of zymase enzyme.

Disaccharides

→ Structure of Sucrose



Glycosidic linkage $\rightarrow$ C₁-C₂ or α - β glycosidic linkage



→ Structure of Maltose

α -D-glucose + α -D-glucose

Glycosidic linkage - C₁-C₄ or α -glycosidic

→ Structure of Lactose

β -D-Galactose + β -D-glucose

Glycosidic $\rightarrow$ C₁-C₄ or β -glycosidic linkage

Poly Saccharides

→ Cellulose

cellulose $\xrightarrow{H_2O^+}$ n β -D-glucopyranose
(plant cell wall) (C₁-C₄ glycosidic linkage)

→ starch $\rightarrow$ Polymer of α -D-glucose

Starch $\xrightarrow{H_2O^+}$ n α -D-glucopyranose
(C₁-C₄ glycosidic linkage)

types:-

Amylose

Amylopectin

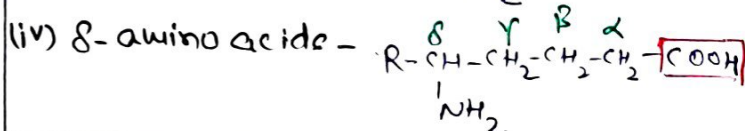
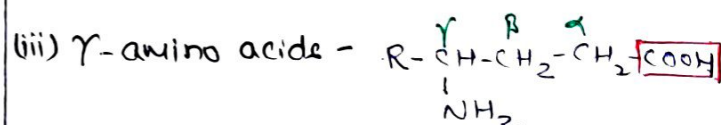
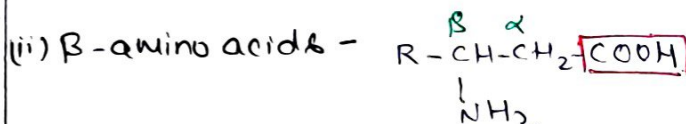
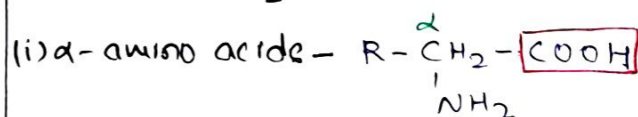
- ① Linear polymer
- ② 15-20% starch
- ③ water soluble
- ④ Glycosidic linkage at C₁ & C₄.

- ① Branched polymer
- ② 80-85% starch
- ③ H₂O insoluble
- ④ linkage at C₁ & C₄ & branched at C₆.

Carbohydrate	Monosaccharide	glycosidic linkage	HA / T F / RS
Sucrose	α -D-glucopyranose + β -D-fructofuranose	C ₁ -C ₂ or α - β	X
Maltose	α -D-glucopyranose + α -D-glucopyranose	C ₁ -C ₄ or α -	✓
Lactose	β -D-galactopyranose + β -D-glucopyranose	C ₁ -C ₄ or β -	✓
cellulose	n β -D-glucopyranose	C ₁ -C ₄ or β -	X
Starch	n α -D-glucopyranose	C ₁ -C ₄ or α -	X
Glycogen	n α -D-glucopyranose	All C ₁ -C ₄ & C ₁ -C ₆	X

AMINO ACID

- α -amino acid is the
- protein is a polymer of α -Amino acid
- All amino acid belongs 'L' family.
- There are 23-natural occurring α -amino acids.
- Glycine, $\text{CH}_2\text{-COOH} \rightarrow$ optically inactive
 NH_2



Amino Acids (α -Amino Acids)

- Based on R

(i) If R contains COOH

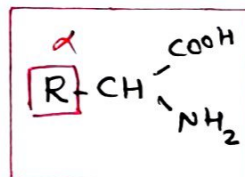
$\rightarrow$ Acidic AA

(ii) If R contains NH_2

$\rightarrow$ Basic AA

(iii) If R neither contain COOH nor NH_2

$\rightarrow$ Neutral



★ Important α -amino acids

Symbol	Name	Structure
Tyr	Tyrosine	HO-  -CH ₂ -CH(NH ₂)-COOH <i>Common</i>
Ser	Serine	HO-CH ₂ -
Gly	Glycine	H-
Ala	Alanine	Me-
Phe [*]	Phenylalanine	Ph-CH ₂ -
Val [*]	Valine	Me ₂ CH-
Leu [*]	Leucine	Me ₂ CH-CH ₂ -
Ile [*]	Isoleucine	Et-CH(Me)-
Lys [*]	Lysine	H ₂ N-(CH ₂) ₄ -
Arg [*]	Arginine	H ₂ N-C(=NH)-NH-(CH ₂) ₃ -
ASP	Aspartic acid	HOOC-CH ₂ -
GLU	Glutamic acid	HOOC-CH ₂ -CH ₂ -

★ TRICK:-

(Neutral)

Tere (Tyr)

Sehar (Ser)

Gaw (Gly)

Ala (Ala)

Phen (Phe)

Vala (Val)

lene (Leu)

Aaye (Ile)

L Lysine B
A Aspartic Acid A
G Glutamic Acid A
AN Arginine B

★ → Essential Amino acid

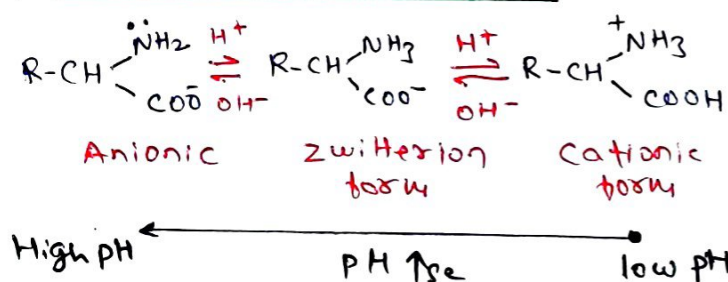
A → Acid

B → Base

① **Essential Amino Acid** → which doesn't synthesize in human body, taken from outside from diet

② **Non-Essential Amino Acid** → which are synthesize in human body.

Zwitter ion:- (Bipolar ion)



Trick → AAJ Katreena hill station par h

A Z cat

H I L par hain.

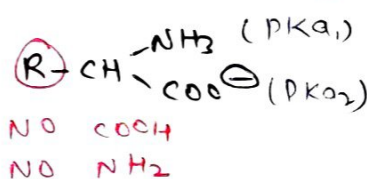
Isoelectric point (pH or pI)

↳ The point where no net migration of ions takes place.

or

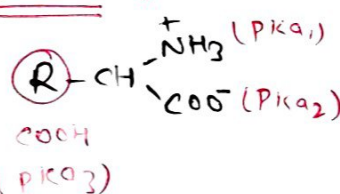
↳ The point where zwitter ion exists (net charge = 0)

Case I - for neutral Amino Acid (usually 6)



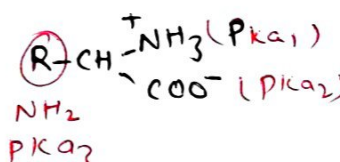
$$\text{pH} = \frac{\text{pKa}_1 + \text{pKa}_2}{2}$$

Case II - for Acidic Amino Acid (usually 2)



$$\text{pH} = \frac{\text{pKa}_2 + \text{pKa}_3}{2}$$

Case III - for Basic AA (usually 10)



$$\text{pH} = \frac{\text{pKa}_1 + \text{pKa}_3}{2}$$

★ Imp. points of α -Amino Acids

① Most natural occurring amino acids belong to 'L' family.

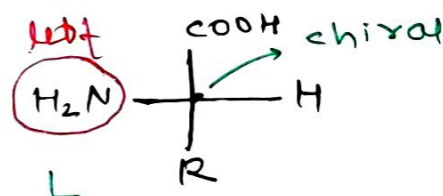
② Glycine is only optically inactive

③ High N.P & B.P

④ water soluble

⑤ Behave like salt

⑥ α -amino acid is monomer of protein.



PROTEIN

• Polymer of α -Amino Acids.

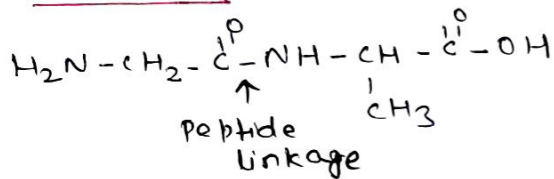
* **Peptide** $\rightarrow$ when α -amino acid on condensation gives peptide.

Linkage $\rightarrow$ peptide linkage (bond)

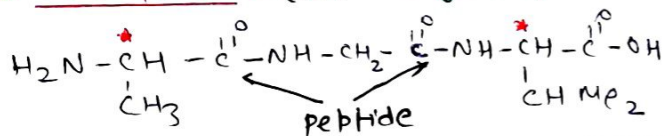
- All amide link are pep. link (x)
- All pep. link are amide link (✓)

eg $\rightarrow$ Dipeptide (Ala-gly)
 tripeptide (Ala-gly-val)
 Polypeptide

Dipeptide $\rightarrow$ (Gly + Ala)



Tripeptide $\rightarrow$ (Ala-gly-val)



Fibrous protein

- Linear
- H_2O insoluble
- H-bonding, Sulphide linkage

eg $\rightarrow$

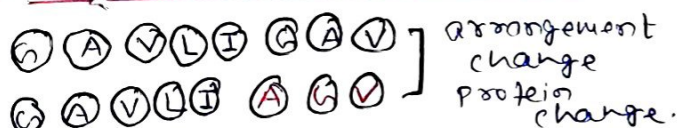
- 1 Keratin (hair)
- 2 Myosin (Muscle)

Globular protein

- Coiled $\rightarrow$ (3D)
- H_2O soluble
- H-bonding, sulphide linkage, dipole-dipole, vanderwaal force of attraction.

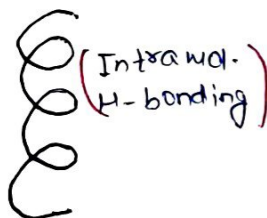
eg $\rightarrow$ Insulin
 Albumin

1 Primary protein structure

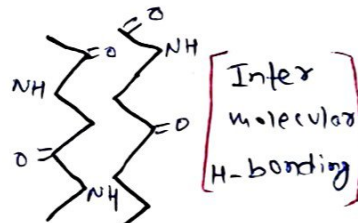


2 Secondary protein structure

α -helix

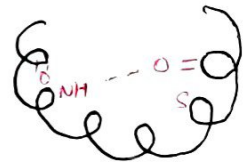


β -pleated



3 Tertiary protein structure

- i) v. waal force
- ii) Dipole-Dipole
- iii) H-bonding
- iv) sulphide linkage



4 Quaternary structure

- * Combination of 2 or more sec. 2 text. structure

* Denaturation of proteins

- protein present in its native form
- protein have its own biological buf.
- Physical or chemical change like temp., H-bonding.
- During denaturation 2° & 3° structure are destroyed but 1° structure remains intact.

eg $\rightarrow$ coagulation of egg white
 curdling of milk.

ENZYMES

- 2° subix $\rightarrow$ **ase** $\rightarrow$ Activation energy $\downarrow$
- Biocatalyst $\rightarrow$ Rate of rxn $\uparrow$
- Globular protein

* **oxidoreductase** $\rightarrow$ behaves as oxidising as well as reducing.

urea $\xrightarrow{\text{urease}}$ Ammonia & CO_2

Maltose $\xrightarrow{\text{Maltase}}$ Glucose

Sucrose $\xrightarrow{\text{Invertase}}$ G + F

starch $\xrightarrow{\text{Amylase}}$ maltose

proteins $\xrightarrow{\text{Trypsin}}$ Amino acids

Ascorbic Acid $\xrightarrow{\text{Ascorbic acid oxidase}}$ Dehydroascorbic acid.

VITAMINS

- * Deficiency cause disease
- * Required in small amount in our diet.
- * Required for optimum growth & perform some biological function.

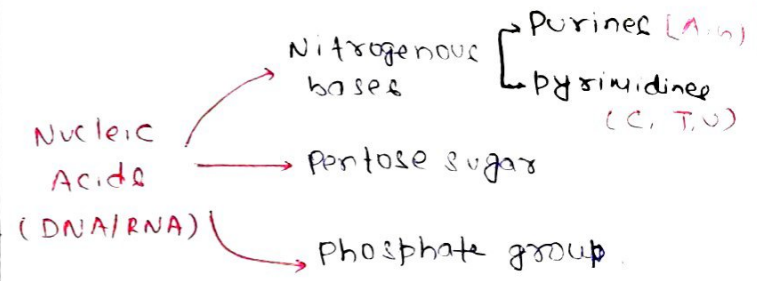
① Fat soluble → vitamins (K & D)

② water soluble → (B & C)

Name of vitamin	Deficiency disease
① Vitamin A (Retinol)	xerophthalmia (hardening of cornea of eye), Night blindness
② Vitamin B ₁ (Thiamine)	Beri beri (loss of appetite, retarded growth)
③ Vitamin B ₂ (Riboflavin)	Cheriosis (bleeding at corners of mouth & lips), digestive disorders & burning sensation of the skin).
④ Vitamin B ₆ (pyridoxine)	convulsions
⑤ Vitamin B ₁₂	Pernicious anaemia (RBC deficient in Hb).
⑥ Vitamin C (Ascorbic acid)	Scurvy (bleeding gums)
⑦ Vitamin D	Rickets (bone deformities in children) & osteomalacia (soft bones & joint pain in adults).
⑧ Vitamin E	Increased fragility of RBC & muscular weakness
⑨ Vitamin K	Increased blood clotting time.

Nucleic Acids

DNA	RNA
① Deoxyribonucleic Acid	① Ribonucleic Acid
② Sugar → β-D-2-deoxyribose	② Sugar → β-D-ribose
③ Base → C, G, A, T	③ Base → C, G, A, U
④ Controlling	④ Synthesize protein
⑤ Double strand	⑤ single strand.



- * **Nucleoside** - Base + Sugar
- * **Nucleotide** - Base + sugar + phosphate grp.
- **Nucleic Acid** → Many nucleotides.

Important points:-

- The two strands are joined by 3'-5' phosphodiester bonds.
- In both chains in blw A & T, 2-H-bond (A = = T) are present, while in C & G, 3H-bonds (C ≡ ≡ G) are present.
- A always attach with T, while C - G.



Biological functions of Nucleic Acids

- DNA is reserve of genetic info & responsible for heredity transmission & exclusively responsible for maintaining the identity of diff. species.
- DNA is capable of self duplication during cell division, therefore identical DNA strands are transferred to daughter cells.
- Important function of nucleic acid is protein synthesis. Proteins are synthesized by RNA molecules in the cell but the message for the synthesis of a particular protein is present in DNA.

HORMONES

↳ Hormones are the molecules that act as intercellular messengers & are poured directly in the blood stream by endocrine glands.

* Types of hormones

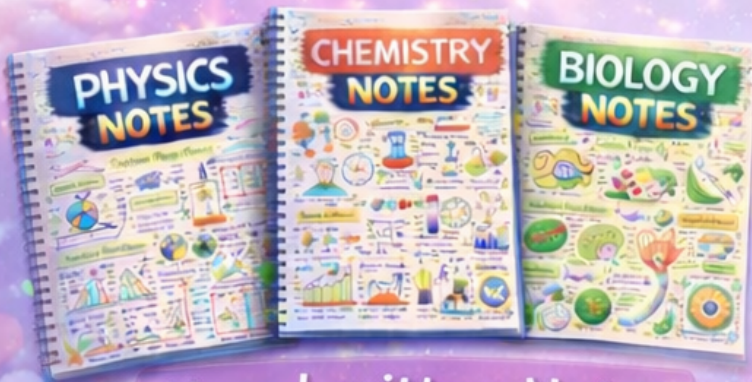
- Steroids:- Estrogens & androgens.
- Polypeptides:- Insulin & endorphins.
- Amino acid derivatives:- Epinephrine & norepinephrine.

Important points:-

- ① They help to maintain the balance of biological activities in the body.
- ② Growth hormones & sex hormones play role in growth & development.
- ③ Hormone insulin helps in keeping the blood glucose level within the limit.
- ④ Glucagon tends to increase the glucose level in the blood. The two hormones together regulate the glucose level in the blood.
- ⑤ Epinephrine & norepinephrine mediate responses to external stimuli.

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